[image] # Tour Ceylon - SafeJourney AI (R26-IT-152)

Tourism Risk and Context Intelligence System: an AI-based tourism risk prediction and advisory platform for Sri Lanka

Final Year Research Project - BSc (Hons) in Information Technology, Department of Information Technology, Sri Lanka Institute of Information Technology (SLIIT), Malabe. Module: IT4010 Research Project.

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1. Project Overview

Tourists in Sri Lanka often plan trips without reliable, combined information on crowding, safety, travel planning and wellness options. Information is scattered across many sources, and help in an emergency can be slow to reach.

Tour Ceylon brings four AI-supported components into one platform: a crowd and risk prediction service, an emergency (SOS) service, an itinerary planner and a wellness recommender. Together they help tourists choose where and when to travel, plan a trip, find wellness options and get help quickly when something goes wrong.

2. Research Objectives

Main objective: To design and build an integrated, AI-based advisory platform that improves the safety, planning and experience of tourists in Sri Lanka.

Sub-objectives (one per component):

- Crowd and risk prediction: predict crowd levels and risk for tourist sites across Sri Lanka and validate the model on unseen sites.
- SOS: provide one-tap emergency reporting with location capture and routing to the nearest responding station.

- Itinerary: generate day-by-day travel plans from the traveller's mood, budget and selected attractions.
- Wellness: recommend Ayurveda and spiritual/meditation centres matched to the traveller's needs.

3. Team and Supervisors

Member	Student ID	Component
De Silva D.S.K (group leader)	IT22108654	SOS and emergency response
Wanniarachchi P.M.R	IT22225092	Itinerary planning
De Silva D.C.M	IT22108586	Wellness / Ayurveda recommendation
Abinaya Rajasekara	IT22090058	Crowd and risk prediction

- **Supervisor:** Mr. Buddika Harshanath
- **Co-supervisor:** Prof. Samantha Rajapaksha

4. System Architecture

The platform has a React front end for each component and Flask back ends. A shared Flask back end (port 5002) serves the travel, SOS, wellness and risk APIs from one PostgreSQL database. The crowd and risk component can also run on its own (port 5000).

5. Components

- **Crowd and risk prediction:** Random Forest pipeline trained on a 197-site dataset. Evaluated with site-holdout (primary), random-split and cross-validation results, reported separately and labelled by split.

- **SOS:** one-tap SOS with GPS capture, incident triage, routing to the nearest station, officer dashboards and an offline queue.
- **Itinerary:** step-by-step wizard (attractions, budget, accommodation, itinerary, recommendations, export) with rule-based scheduling and PDF export.
- **Wellness:** dual-path wizard (Ayurveda and spiritual/meditation) with a dosha classifier and a weighted matching score.

6. Tech Stack

- **Back end:** Python, Flask, Flask-SocketIO
- **ML/AI:** scikit-learn (Random Forest), pandas, NumPy
- **Front end:** React, Vite
- **Database:** PostgreSQL, SQLite
- **Other:** JWT authentication, WeasyPrint / ReportLab (PDF export)

7. Folder Structure

```
R26-IT152/
  backend/
    data/
    models/
    outputs/
    scripts/
  components/
    scripts/
    shared/
  frontend/
    src/
  tourism_risk_system (1)/
    tourism_risk_system/
  tourism-unified/
    public/
    scripts/
    src/
```

```
README.md
```

8. Setup and Installation

Requirements: Python 3.10+, Node.js 18+, PostgreSQL, Git.

```
git clone https://github.com/SavinDeSilva/R26-IT152.git
cd R26-IT152
python -m venv .venv
.venv\Scripts\activate
```

Install the Python packages for each component you want to run:

```
pip install -r components\<component>\backend\requirement
```

Configuration: create a `.env` file in `components\shared\backend` with your own `DATABASE_URL` and secret keys. The `.env` file is not stored in the repository.

9. How to Run

Shared back end (port 5002):

```
cd components\shared\backend
python run.py
```

Check it at `http://127.0.0.1:5002/api/health`.

Crowd and risk back end on its own (port 5000):

```
cd components\risk-management\backend
python app.py
```

Front ends (run `npm install` once, then `npm run dev` inside each folder):

Front end	Folder	Port
SOS (tourist)	components\sos\frontend\tourist	5175
SOS (police)	components\sos\frontend\police	5176
Itinerary	components\itinerary\frontend	5180
Wellness	components\wellness\frontend	5181
Crowd and risk	components\risk-management\frontend	5182

10. Project Status

Final-year research project, submission stage. All four components are implemented and run on the shared back end. Known limitations are documented in the individual theses and the group report.

11. License

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